OpenTRE

Open Trusted Research Ecosystem

BLOG POST

The Catalogue and the Crisis

Why the UK's £600 Million Investment Needs Protocols, Not Another Platform

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OPENTRE

This is part of a series of papers arguing that the UK's health data infrastructure should build protocols, not services. Companion papers covering the full technical case, a visual executive summary, the political and rights-based vision, an enforceable rights framework, a protocol specification, and an implementation roadmap are available at opentre.org. For the US perspective, see *The Current Wars of Biomedical Data*.

IN 60 SECONDS

What: OpenTRE proposes that the HDRS invest its £600 million in open interoperability protocols (shared cataloguing rules for health data) rather than another centrally operated platform.

Why: The UK has spent over two decades and billions of pounds on centralised health data programmes (NPfIT, care.data, GPDPR, FDP) that follow the same pattern: centralise, meet resistance, retreat, rebrand. The problem is not technology. It is the absence of shared rules.

What should be done now: Five Bermuda-style mandates (24-hour governance disclosure, federated-first presumption, protocol conformance, credential portability, and reciprocal contribution) would give the HDRS a durable foundation that survives political cycles.

▶ **KEY TERMS & ACRONYMS**

THE CATALOGUE AND THE CRISIS

One set of rules, written once, connecting every institution

A clinical researcher at University College London is studying antibiotic resistance patterns across three regions of England. The data she needs exists in three separate Secure Data Environments. She has ethics approval. She has funding. She has a clear research question. What she does not have is a way to query across all three environments without negotiating a separate data access agreement with each one, a process that will take, by her own estimate, between nine and fourteen months per SDE[?]. By the time she can run the analysis, her grant will have expired. This is exactly the kind of problem OpenSAFELY solved during COVID-19: code travelling to 58 million GP records rather than records travelling to researchers. The architecture works. The governance interoperability does not yet exist.

This is not a technology problem. It is a cataloguing problem. The data exists. The shared rules for accessing it do not. The same problem existed in the library world before 1841, when a political refugee published the rules that eventually connected every library on earth.

WHY THIS MATTERS FOR THE HDRS

- **The pattern is identical:** before Panizzi, every library was an island with its own catalogue. Today, every NHS SDE is an island with its own governance process. The £600 million HDRS can write the shared rules, or build another island.

- **Metadata travels, content stays:** Panizzi's rules standardised how to describe a book, not the book itself. OpenTRE standardises how to describe data access conditions, not the patient data. The "book" never leaves the shelf.
- **Sovereignty is preserved:** no library surrendered its collection. No NHS Trust needs to surrender its data. Only the catalogue entries (the governance interfaces) need to conform.

In 1823, a young Italian revolutionary named Antonio Panizzi arrived in England. He had been sentenced to death in absentia for conspiracy against the Duke of Modena. He spoke little English. He had no money, no connections, and no prospects. Within fourteen years he would become Keeper of Printed Books at the British Museum Library, and in 1841 he would publish 91 cataloguing rules that became the foundation of every library interoperability standard that followed.

Before Panizzi, every library was an island. Each institution catalogued its holdings in its own way, using its own conventions, its own entry formats, its own filing logic. A scholar who could navigate the Bodleian's catalogue was lost in the British Museum's. Finding a book meant knowing the local system. There was no shared language for describing what a library held.

Panizzi's insight was that the problem was not the buildings or the collections. The problem was the absence of shared rules. His 91 rules specified how to record an author's name, how to handle anonymous works, how to cross-reference editions, how to standardise titles. They were, in modern terms, a protocol: a set of agreements about representation that allowed any two institutions to understand each other's holdings without merging their collections.

The rules did not require libraries to reorganise their shelves, rebuild their reading rooms, or surrender their collections to a central repository. Each library kept its own governance, its own physical arrangements, its own traditions. Only the catalogue entries (the metadata describing what was held and how to find it) needed to conform to a shared standard. The books themselves never left the shelf. A scholar could discover what existed anywhere; the sensitive content stayed where it belonged. In OpenTRE's terms: the catalogue (governance metadata) travels; the patient data (the book) never does.

That lineage is direct and unbroken. Panizzi's rules (1841) led to Charles Ammi Cutter's *Rules for a Dictionary Catalog* (1876) and, in the same year, Melvil Dewey's Decimal Classification, which gave every library a shared language for organising knowledge by subject. Cutter standardised how to describe a book; Dewey standardised where to put it. Together they solved both halves of the interoperability problem. That dual tradition led to the Anglo-American Cataloguing Rules (1967) and to Henriette Avram's MARC format at the Library of Congress (1966), which gave cataloguing a machine-readable encoding. MARC led to Z39.50, the federated query protocol that lets a researcher in London search the catalogue of a library in Tokyo without either institution surrendering its data. Z39.50 led to WorldCat, a federated discovery catalogue spanning tens of thousands of libraries across 170 countries. And the entire system rests on persistent identifiers like ISBN and ISSN, which let any catalogue entry point unambiguously to the same work.

A political refugee wrote the rules that let every library in the world talk to each other. No central warehouse was required. No single institution controlled the system. The protocol was the agreement. Everything else was implementation.

I find myself thinking about Panizzi's rules as I watch the UK prepare to spend £600 million on a Health Data Research Service. The NHS has a cataloguing problem. Institutions that hold health data are essentially running incompatible catalogues: bespoke governance agreements, variable access processes, months-long negotiations, each request treated as a unique transaction. The data exists. The shared rules do not.

Panizzi built the Round Reading Room and expanded the collection, but his lasting contribution was the 91 rules that made every catalogue interoperable. The UK's health data infrastructure needs the same kind of reform.

The Health Data Research Gateway, which we built at HDR UK, was the first metadata catalogue of health research datasets across all four UK nations. We also began aligning governance processes by developing a unified Data Access Request process using the Five Safes framework. Without a political mandate, however, uptake has been hard. The challenge of interoperable governance remains.

> **KEY TAKEAWAY**
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> The NHS has a cataloguing problem, not a technology problem. Shared rules for describing data access conditions, not more buildings, are what connect institutions.

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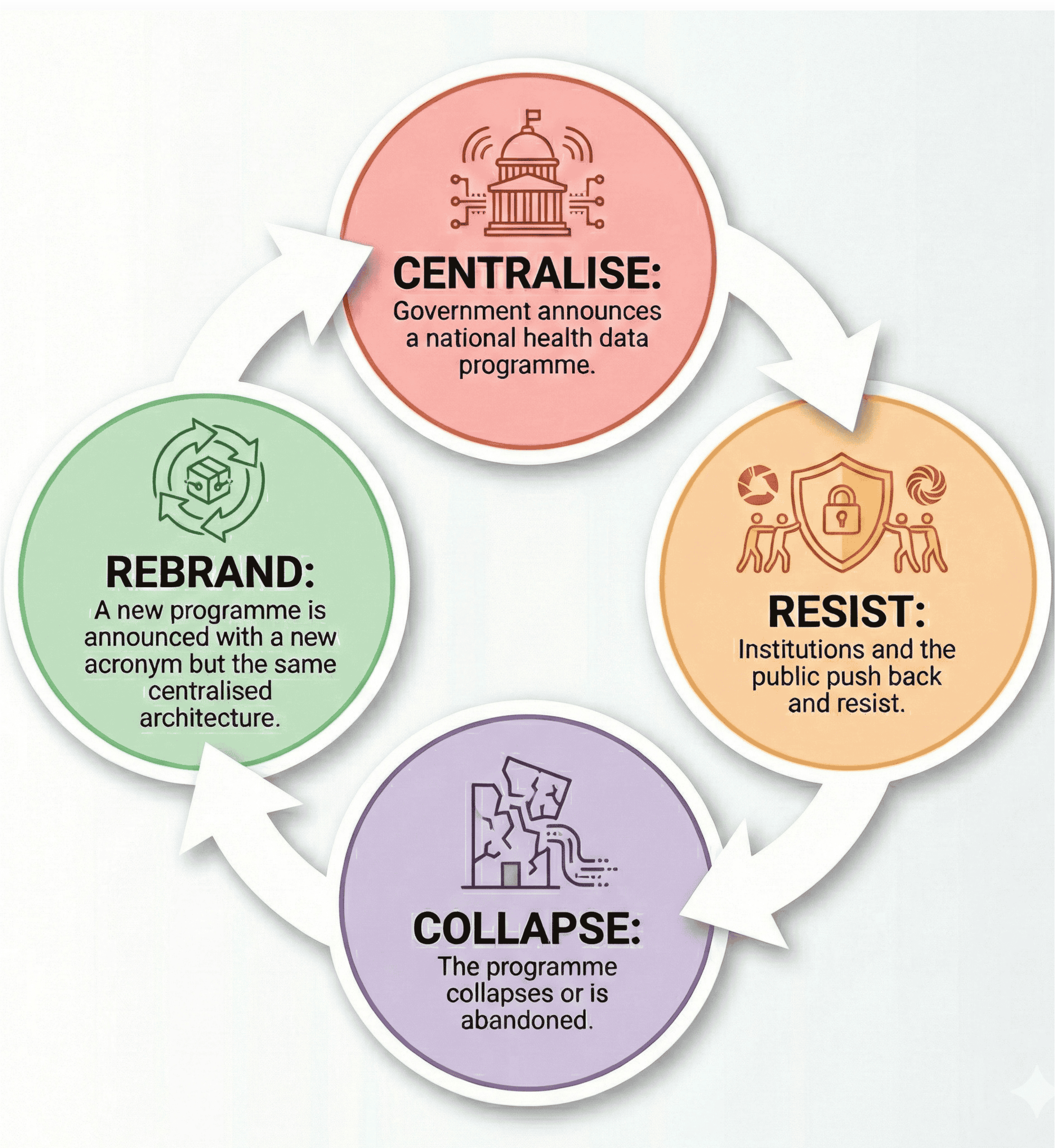
The Pattern That Keeps Repeating



Every major UK health data programme has tried to build the British Library. None of them thought to write the cataloguing rules first.

The **National Programme for IT** (2002-2011) tried to build a single electronic health record system for the entire NHS. Cost: £12.7 billion. Result: largely abandoned. Twelve billion pounds, and the scale of that failure should have been instructive. The Public Accounts Committee's verdict: "an overambitious and unwieldy centralised model." It was the equivalent of demanding that every hospital in England physically relocate its records to a single building. The British Library's St Pancras building, by comparison, cost £511 million and took 35 years from conception to completion (1962-1997). NPfIT tried to build something vastly more ambitious in a fraction of the time.

Care.data (2013-2016) tried to extract GP data into a central repository. It cost the taxpayer £8 million. 1.5 million patients opted out. Programme cancelled. The library equivalent: photocopying every book in every local library and shipping the copies to a central warehouse, without asking the borrowers.

GPDPR (2021) tried the same thing with a different acronym. Within weeks of announcement, opt-outs nearly doubled (June-July 2021). 1.27 million additional people refused in a single month. Programme paused indefinitely.

The **GPES Data for Consented Research Directions 2026** represent the latest iteration. On 10 February 2026, the Secretary of State directed NHS England to share GP health records with approved research studies where participants have given explicit consent. The consent model is a real step forward. But look beneath it: the same GDPPR (GPES Data for Pandemic Planning and Research) data pipeline, originally justified under pandemic emergency powers, repurposed for a new purpose. Data still flows from GP systems to NHS England to study platforms. The architecture remains centralised even when the consent is not. OpenTRE's protocol layer could make this Directions work better: machine-readable consent verification, automated Type 1 objection enforcement, and interoperable SDE access. The result would be auditable, computable governance in place of a manual multi-stage approval process.

The **Federated Data Platform** (2023-present), despite its name, is a centrally mandated system under a £330 million single-vendor contract. Think of it as building a grand reading room with a single proprietary catalogue system: excellent for patrons within its walls, and useful for direct care analytics within participating trusts. But a single reading room cannot serve every community. Several trusts with mature local capabilities have concluded it would reduce their functionality. Greater Manchester, which built its own interoperable platform, declined entirely.

The FDP is a useful node, one that needs protocol interfaces to participate in a wider ecosystem.

And then there is the British Library itself. In October 2023, a Rhysida ransomware attack took its digital services offline for months and cost an estimated £7 million. Everything was concentrated in one infrastructure; when it fell, everything fell with it. The physical inter-library loan network, federated by design, kept working. The lesson is hard to miss.

Federation distributes risk rather than eliminating it. A protocol-based ecosystem must address the expanded attack surface through mandatory security conformance testing, incident response coordination across SDEs, and disclosure control at the aggregation layer. The HDRS should fund a shared security operations capability for the SDE Network.

REBUTTAL: "FEDERATION INCREASES POINTS OF FAILURE"

The sceptic: "Every new SDE endpoint is another attack surface. Centralisation is simpler to defend."

The response: Centralisation is simpler to defend until it fails, at which point everything fails simultaneously (see: British Library, October 2023). A federated model limits the blast radius of any single compromise. The HDRS should establish a **Federated Security Council** with a shared Incident Response Protocol. The protocol includes a "circuit breaker" mechanism: if any node is compromised, the network can isolate it within minutes while the rest continues to operate. Spreading risk across many nodes, with a coordinated response plan, beats concentrating it in a single target.

Four programmes. Over two decades. Billions of pounds. Partial successes (the Spine, Summary Care Record) but a recurring pattern: build a bigger library, meet resistance, retreat, rebrand. The UK keeps trying to centralise the collection when what it needs is to standardise the catalogue.

A protocol layer is not a repudiation of past investment. It is an insurance policy against platform obsolescence. The £12.7 billion spent on NPfIT, the £330 million committed to the FDP, the existing SDE Network: none of these need to be abandoned. They need protocol interfaces that protect the public's investment by making each system interoperable with the next, regardless of which vendor built it or which government funded it. The alternative (another generation of proprietary platforms) is the real sunk cost.

There is a middle way between the centralisation that provokes resistance and the fragmentation that prevents research. Protocol-based federation preserves NHS Trust sovereignty (every Trust keeps its data, its governance, its Caldicott Guardian's authority) while enabling national-scale queries across the entire SDE Network. Local control and national reach are not opposites. They are two sides of the same protocol. For UK policymakers, this is the win-win: Trusts retain the autonomy they will fight to protect, while HDRS delivers the cross-institutional research capability that justifies the £600 million investment.

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